BFKL Terms (corrected)

Result

Red: remove. Blue: add. Every blue term is the coefficient of $\log\frac{\vee}{k^+}$ in a logarithm of the same row of `Finite_terms_1.tex` that contains $\vee$ and that this list had left out (the notes after each row say which one). Every row of this list was checked against $\partial/\partial\log\vee$ of the same row of `Finite_terms_1.tex` at $\frac{\vee}{k^+}=10^7\dots10^9$: agreement to 3×10^{-6} or better (`corrected_check.json`).

Changes.

- Group I Row II part 1: coefficient 2, not 1.
- Group I Row III: d^3N_1 replaced; six terms added to d^3N_2.
- Group I Row IV, Group II Row I part 1, Group III Rows I, II: the $\mathbb{T}^{ij}$ term added.
- Group II Row I part 2: 2d, 2e, 2f removed (they add up to zero); 2b, 2c added.
- Group II Row IV, Group III Row III (both parts), Rows V, VI: marked NA, but they contain $\log\frac{\vee}{k^+}$; added.
- Group I Row II 2a, Row V (all parts): correct.

What the corrected list gives.

1. **Three ρ's** cancel after symmetrization (checked: `LogVeeKplus_BFKL.pdf`, all 39 structures; `BFKL_terms_check.pdf`, Sec. 5). Without the blue $\mathbb{T}^{ij}$ terms they cannot: the kernel $\frac{1}{(z-w)^2}$ of your Group I Row IV, Group II Row I part 1 and Group III Rows I, II is not integrable at $z\to w$ (note after Group I Row IV).

2. **Two ρ's** add up to
$$\sum_{\text{Groups I–III}}\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[\mathcal{B}-\mathcal{J}\Big],\tag{0.1}$$
with $\mathcal{B}$ = BFKL⊗LO (your Eq. (LOBFKL)) and $\mathcal{J}$ = JIMWLK⊗LO, the brackets of Secs. 4 and 3 of `Evolution_terms_corrected.tex`.

3. **Why $-\mathcal{J}$.** In every row the soft gluon is integrated over $\Lambda<p^+<\vee$. With T = the integrand for $p^+\ll k^+$ and P = the integrand for $p^+\gg k^+$,
$$\int_\Lambda^\vee\frac{dp^+}{p^+}\,G\Big(\frac{p^+}{k^+}\Big)=T\log\frac{k^+}{\Lambda}+P\log\frac{\vee}{k^+}+\dots=T\log\frac{\vee}{\Lambda}+(P-T)\log\frac{\vee}{k^+}+\dots$$
   - $T\log\frac{\vee}{\Lambda}$ is in your $\log\frac{\vee}{\Lambda}$ list, and $\sum T=\mathcal{J}$;
   - $\sum P=\mathcal{B}$;
   - so this list is $\sum(P-T)=\mathcal{B}-\mathcal{J}$.

4. **Both lists together:**
$$\boxed{\log\Big[\frac{\vee}{\Lambda}\Big]\mathcal{J}+\log\Big[\frac{\vee}{k^+}\Big](\mathcal{B}-\mathcal{J})=\log\Big[\frac{k^+}{\Lambda}\Big]\mathcal{J}+\log\Big[\frac{\vee}{k^+}\Big]\mathcal{B}.}\tag{0.2}$$
   BFKL comes only with $\log\frac{\vee}{k^+}$, and JIMWLK only with $\log\frac{k^+}{\Lambda}$, as I said before. The $-\mathcal{J}$ of this list is not a new claim: your JIMWLK terms are written with $\log\frac{\vee}{\Lambda}=\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}$, and the $-\mathcal{J}\log\frac{\vee}{k^+}$ here removes the second piece.

5. **Group IV** (Sec. 4) moves $\log\frac{\vee}{k^+}\,\mathcal{J}$ from your $\log\frac{\vee}{\Lambda}$ list to this one. Then this list is exactly your Eq. (LOBFKL) with $\delta Y=\log\frac{\vee}{k^+}$ (Sec. 5).

6. **At mid rapidity**, $k^+=\sqrt{\vee\Lambda}$: $\log\frac{\vee}{k^+}=\log\frac{k^+}{\Lambda}=\frac12\log\frac{\vee}{\Lambda}$. So Eq. (0.2) becomes $\frac12\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})$, the form in `Evolution_terms_corrected.tex`. It is the same equation.

1 Group I (with c.c.)

1.1 Row I

NA

1.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\,1{\times}2{\times}\log\left[\frac{\vee}{k^+}\right]\int_{w,w',x,x'}e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.$$
$$\left.\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right]\right. \tag{1.1}$$

$$\frac{d^3N_{2\,a}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{hbc}}{32\pi^4}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]\times$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]$$
$$+\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{abc}}{32\pi^4}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2}\right.$$
$$\left.-\frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right] \tag{1.2}$$

Row II. Part 1: in `Finite_terms_1.tex` the pole bracket multiplies $\log\frac{\vee}{k^+}+\log\left(\frac{\vee}{k^+}-1\right)=2\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right)$, so the coefficient is 2. Part 2a: correct (nothing depends on z, so $\int_z$ is dropped).

1.3 Row III

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4\,f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}$$
$$\left.\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\right\} \tag{1.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]$$
$$+\frac{ig^4\,f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x'}\left[\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\right.$$
$$-\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}$$
$$+\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}$$
$$-\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}$$
$$+\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}$$
$$\left.-\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\right]$$
$$\left.\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\right\} \tag{1.4}$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
\end{aligned}
\tag{1.5}
$$

Row III. The $\log\frac{\vee}{k^+}$ of this row also sits in three logarithms of `Finite_terms_1.tex` that you did not count:

$$
\log\left|\frac{\vee\,(x-z)^2 - k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2 - k^+(x-w)^2}\right|, \qquad \log\left|\frac{k^+(x-w)^2 - \vee\,(x-z)^2}{k^+(x-w)^2 - \Lambda\,(x-z)^2}\right| \quad (d^3N_1),
$$

$$
\log\left|\frac{(x-w)^2k^+ - (x-z)^2\vee}{(x-w)^2k^+ - (x-z)^2(k^+ + \Lambda)}\right| \quad (d^3N_2); \qquad \text{each} = \log\frac{\vee}{k^+} + O(1) \text{ for } \vee\to\infty.
$$

In d^3N_1 the first one cancels your term exactly (same kernel, opposite sign); the second one gives the blue term. In d^3N_2 the third one gives the six blue terms. d^3N_{3a} is correct; d^3N_{3b} has no $\vee$.

1.4 Row IV

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \\
&\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{1.6}
$$

Row IV. The blue term is the $\log\frac{\vee}{k^+}$ inside the first logarithm of the row in `Finite_terms_1.tex`:

$$
\log\left[\frac{(\vee - k^+)(x-z)^2 + k^+(x-w)^2}{\Lambda(x-z)^2 + k^+(x-w)^2}\right] = \log\frac{\vee}{k^+} + \log\left[\left(1-\frac{k^+}{\vee}\right)\frac{(x-z)^2}{(x-w)^2} + \frac{k^+}{\vee}\right] + O\Big(\frac{\Lambda}{k^+}\Big).
$$

Together, your term and the blue term have the kernel (with δ^{ij} the $\frac{1}{(z-w)^2}$ parts cancel)

$$
\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\left[\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(z-w)^2(y-z)^2(x-z)^2} - \frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(x-w)^2(y-z)^2(x-z)^2} - \frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2(x-w)^2(y-z)^2(z-w)^2}\right],
$$

which is integrable at $z\to w$. Your term alone, $\propto\frac{1}{(z-w)^2}$, is not: its z integral diverges, so the three-ρ terms could not cancel. The same holds for Group II Row I part 1 and Group III Rows I, II.

1.5 Row V

First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \\
&- \frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]
\end{aligned}
\tag{1.7}
$$

Second part

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\,\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\vee}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^f(y) \tag{1.8}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{1.9}$$

$$\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.$$
$$\left.-\,\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\vee}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y) \tag{1.10}$$

Row V. All parts correct.

2 Group II

2.1 Row I

2.1.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}\\
&\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]\\
&+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\left(-\frac{ig^4}{16\pi^5}\right)\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\\
&\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\\
&\times\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\\
&\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{2.1}
$$

Row I, part 1. The blue terms are the $\log\frac{\vee}{k^+}$ inside $\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\right]$ and its partner with $w\to w'$ (as in Group I Row IV).

2.1.2 Second part

Row I, part 2. Remove 2d, 2e, 2f: in `Finite_terms_1.tex` their explicit $\log\frac{\vee}{k^++\Lambda}$ cancels exactly against the $\log\frac{\vee}{k^+}$ inside $\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\right]$ and $\log\left[\frac{(\vee-k^+)(x'-z)^2+k^+(x'-w')^2}{\Lambda(x'-z)^2+k^+(x'-w')^2}\right]$ in the same rows. The $\log\frac{\vee}{k^+}$ of part 2 comes from 2b and 2c, through the same two logarithms (2a: they cancel between numerator and denominator):

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(x-w)^2\,(x'-w')^2}\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]
\end{aligned}
\tag{2.2}
$$

$$\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(x-z)^2\,(x'-z)^2}$$
$$\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$\left. - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.3}$$

The first term of d^3N_{2c} has the kernel $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$: the measured gluon is emitted by the soft gluon z in the amplitude and in the conjugate amplitude. This is the kernel of the second line of your Eq. (LOBFKL); no term of your list had it.

Remove:

$$\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}$$
$$\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.$$
$$\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.4}$$

$$\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}$$
$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$
$$\left. + \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.5}$$

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\left[\frac{\vee}{k^+}\right]\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}$$
$$\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x') - f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{2.6}$$

2.2 Row II

NA

2.3 Row III

NA

2.4 Row IV

NA

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E - \frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right]\left[U^{e'c}(y') - U^{e'c}(x')\right]\times\left[U^{ec}(y) - U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{e'c}(z) - U^{e'c}(x')\right]\times\left[U^{ec}(z) - U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{2.7}$$

Row IV. Both terms come from the lower end $\xi \to \frac{k^+}{\vee}$ of $\int_{k^+/\vee}^1 d\xi$ in `Finite_terms_1.tex`:

- first term: your subtraction is $\frac{1}{\xi^2}\frac{\xi}{(1-\xi)_+}e^{-ik\cdot(y'-y)\frac{1}{\xi}} = \frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}} - e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}$. With $\frac{1}{\xi(1-\xi)} = \frac{1}{\xi} + \frac{1}{1-\xi}$, the part $-\frac{e^{-ik\cdot(y'-y)}}{\xi}$ gives $-e^{-ik\cdot(y'-y)}\int_{k^+/\vee}^1\frac{d\xi}{\xi} = -e^{-ik\cdot(y'-y)}\log\frac{\vee}{k^+}$; the rest has no $\log\frac{\vee}{k^+}$. Together with Group I Row II part 1 (coefficient 2) and Row V 2b UV, its $\frac{1}{\epsilon}$ cancels: no pole is left in this list.
- second term: the term $\frac{1-\xi}{\xi}$ of P_{gg} at small ξ. There $e^{-ik\cdot(y'-y)/\xi}$ probes momentum $\frac{k}{\xi}$, and $\int d^2(y'-y)\,e^{-iq\cdot(y'-y)}\log(y'-y)^2\,h(y'-y) \to -\frac{4\pi}{q^2}h(0)$ puts $y = y' = z$ and gives a factor ξ^2, so $\int_{k^+/\vee}\frac{d\xi}{\xi} = \log\frac{\vee}{k^+}$. The factor $\frac{4\pi^2}{k^2}$ is written as $\int_{w,w'} e^{-ik\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$. This term is BFKL: the measured gluon is emitted by the gluon at z on both sides.

3 Group III (with c.c.)

3.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\left(-\frac{ig^4}{8\pi^5}\right)\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}$$
$$\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.1}$$

3.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\left(-\frac{ig^4}{8\pi^5}\right)\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}$$
$$\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.2}$$

Rows I, II. The blue terms are the log $\frac{\vee}{k^+}$ inside log $\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x-w)^2+\Lambda(x-z)^2}\right]$, as in Group I Row IV.

3.3 Row III

3.3.1 First part

NA

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{3.3}$$

3.3.2 Second part

NA

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.4}$$

3.4 Row IV

NA

3.5 Row V

NA

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right] \tag{3.5}$$

3.6 Row VI

NA

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\Big[\frac{\vee}{k^+}\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big] \tag{3.6}$$

Group III Rows III, V, VI. The blue terms come from the $\Big[\frac{1}{p^+}\Big]_+\delta_{jm}\delta_{ik'}$ term of `Finite_terms_1.tex`:

$$\int_0^{\vee-k^+}dp^+\Big[\frac{1}{p^+}\Big]_+ e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} = -\Big[\log\frac{\vee}{k^+}+\log\Big(1-\frac{k^+}{\vee}\Big)+\gamma_E+\log\big(ik\cdot(y'-z)\big)+E_1\Big(ik\cdot(y'-z)\frac{\vee-k^+}{k^+}\Big)\Big]$$

(the same way in part 2, where $\frac{1}{p^+(x-z)^2+k^+(x-w)^2}\to\frac{1}{k^+(x-w)^2}$ at $p^+=0$). The kernels are those of the $\log(1-\frac{k^+}{\vee})$ lines of the same rows of `Finite_terms_1.tex`, with the opposite sign. The other terms of these rows give no $\log\frac{\vee}{k^+}$ (the oscillating E_1, $e^{-ik\cdot(y'-z)(\vee-k^+)/k^+}$ terms vanish after $\int_{y'}$).

4 Group IV: the $\log\frac{\vee}{k^+}$ part of your $\log\frac{\vee}{\Lambda}$ list

Your $\log\frac{\vee}{\Lambda}$ terms add up to $\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\big[\frac{\vee}{\Lambda}\big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\,\mathcal{J}$ (`Evolution_terms_corrected.tex`, Sec. 3). With $\log\frac{\vee}{\Lambda}=\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}$:

- $\log\frac{k^+}{\Lambda}\,\mathcal{J}$ stays in the evolution list: JIMWLK, the evolution of the target from Λ up to k^+;
- $\log\frac{\vee}{k^+}\,\mathcal{J}$ belongs to this list. In colour-singlet form (Eq. (3.4) of `Evolution_terms_corrected.tex`, with $\log\frac{\vee}{\Lambda}\to\log\frac{\vee}{k^+}$; $K(a,b,z)=\mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$):

$$\frac{d^3N_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\Big[\frac{\vee}{k^+}\Big]\frac{1}{N_c^2-1}\int_{x,y,z,w,w'}e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\Big[$$
$$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[4N_c{\rm Tr}\{U^\dagger(w)U(w')\}$$
$$-2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}-2{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-4N_c{\rm Tr}\{U^\dagger(x)U(w')\}$$
$$+2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}+2{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[4N_c{\rm Tr}\{U^\dagger(x)U(y)\}$$
$$-2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}-2{\rm Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-4N_c{\rm Tr}\{U^\dagger(w)U(y)\}$$
$$+2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}+2{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-2N_c{\rm Tr}\{U^\dagger(w)U(w')\}+2N_c{\rm Tr}\{U^\dagger(x)U(w')\}$$
$$-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(w')U(z)\}-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}$$
$$+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w')\}$$
$$+{\rm Tr}\{T^eU^\dagger(w')U(x)\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w')U(z)\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-2N_c{\rm Tr}\{U^\dagger(w)U(w')\}+2N_c{\rm Tr}\{U^\dagger(w)U(y)\}$$
$$+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w)U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}$$
$$-{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(w)U(z)\}-{\rm Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}$$
$$-{\rm Tr}\{T^eU^\dagger(w)U(w')\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\}+{\rm Tr}\{T^eU^\dagger(w)U(y)\}\,{\rm Tr}\{T^eU^\dagger(w)U(z)\}\Big]$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[2N_c{\rm Tr}\{U^\dagger(x)U(w')\}-2N_c{\rm Tr}\{U^\dagger(x)U(y)\}$$
$$-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}-{\rm Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}$$

$$
\begin{aligned}
&+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(x)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\\
&-\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}-\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}\\
&+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(y)\}\\
&-\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\Big]
\end{aligned}
\tag{4.1}
$$

5 Total of this list

Groups I–III give $\log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J})$, Group IV gives $\log\frac{\vee}{k^+}\,\mathcal{J}$. Their sum is $\log\frac{\vee}{k^+}\,\mathcal{B}$. In colour-singlet form this is your Eq. (LOBFKL) with $\delta Y=\log\frac{\vee}{k^+}$ (same normalization as your App. E):

$$
\begin{aligned}
\frac{dN}{d^2w\,d^2w'\,dy}=-\frac{\alpha_s^2N_c\delta Y\log\left[\frac{\vee}{k^+}\right]}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle\\
\Big[\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}\\
-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\right\}\\
+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')(2K(x,y,z)-K(x,x,z))\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\right\}\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')(2K(x,y,z)-K(y,y,z))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\right\}\Big]
\end{aligned}
\tag{5.1}
$$

and the evolution list keeps

$$
\left.\frac{d^3N}{d^3k}\right|_{\mathrm{JIMWLK}}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\,\mathcal{J}.
\tag{5.2}
$$

The physical picture: the soft gluon p^+ has $\Lambda<p^+<\vee$. For $p^+>k^+$ it is part of the projectile and can emit the measured gluon (BFKL, interval $\log\frac{\vee}{k^+}$); for $p^+<k^+$ it is emitted next to the target (JIMWLK, interval $\log\frac{k^+}{\Lambda}$). Each soft gluon is counted once.